

The Energy Hinge

One conservation principle, three different physical ledgers

Working rule: Choose a system boundary, name the stores and transfers, assign signs, and then apply conservation. The conservation law balances the ledger; a separate physical relation usually tells you what belongs in it.

Name: _____

Class: _____

Companion task to *The Placeholder Force*

1. Kirchhoff’s second law: energy per charge

An ideal 12.0 V battery is connected in one loop to $R_1 = 4.0\,\Omega$ and $R_2 = 2.0\,\Omega$ in series. The connecting wires have negligible resistance.

A. Draw the ledger

Sketch the loop and choose a current direction. Mark the battery as an energy rise per coulomb and each resistor as an energy fall per coulomb. State what system you are tracking.

B. Apply the hinge

Begin with the closed-loop energy ledger

$$+\mathcal{E} - V_1 - V_2 = 0.$$

Only then use the empirical resistor relation $V = IR$ to find I , V_1 and V_2 . Show that the signed voltage changes sum to zero.

C. Restore the word “energy”

A charge of 3.0 C passes through the loop. Calculate the energy supplied by the battery and the energy transferred in each resistor. Explain why voltage is an energy-per-charge quantity rather than an energy store.

D. Interrogate conservation

What does Kirchhoff’s loop rule tell you? What does it *not* tell you that Ohm’s law or a model of the battery must supply?

2. First law: heat, work and internal energy

A sample of 0.20 mol of monatomic ideal gas absorbs $Q = 500\,\text{J}$ while expanding and doing $W = 200\,\text{J}$ of work on its surroundings. Use $R = 8.31\,\text{J mol}^{-1}\,\text{K}^{-1}$.

A. Choose the boundary

Draw the gas as the system. Add arrows for energy transferred by heating and by mechanical work. State which transfer is into the system and which is out.

B. Apply the hinge

Using the sign convention that $Q > 0$ enters the gas and $W > 0$ is done by the gas, write

$$\Delta U = Q - W.$$

Find the change in internal energy and express the same calculation as a signed zero-sum ledger.

C. Add physical closure

For a monatomic ideal gas,

$$\Delta U = \frac{3}{2}nR\Delta T.$$

Find ΔT . Identify which part of your reasoning is universal conservation and which part depends on a particular model of matter.

D. Interrogate the terms

Why are Q and W transfers rather than quantities contained in the gas? Would the first law alone determine the final pressure or volume?

3. Nuclear binding: where did the mass go?

A proton and neutron initially far apart form a deuteron. Use

$$m_p = 1.007276\,\text{u}, \quad m_n = 1.008665\,\text{u},$$

$$m_d = 2.013553\,\text{u}, \quad 1\,\text{u}\,c^2 = 931.5\,\text{MeV}.$$

The released energy appears as radiation and recoil kinetic energy.

A. Draw the system change

Represent the separated nucleons, the bound deuteron and the outgoing energy. Draw two possible system boundaries: nucleons only; nucleons plus emitted radiation.

B. Apply the hinge

Calculate the mass defect

$$\Delta m = m_p + m_n - m_d$$

and hence the total released energy $E_{\text{rel}} = \Delta m\,c^2$. Write a complete initial-energy = final-energy ledger.

C. Interpret binding energy

Explain why the deuteron’s rest mass is lower than the sum of the separated masses. Why is the binding energy quoted as a positive number even though the bound state has lower energy? What would be required to separate it again?

D. Interrogate conservation

Does conservation predict the value of the binding energy? Identify the empirical input that makes the calculation possible.

Synthesis: conservation balances a ledger; it does not write one

1. Complete the comparison.

Case	Conserved ledger	Extra relation or empirical input	What crosses the boundary?
Circuit			
Gas			
Deuteron			

2. Develop the central claim.

Respond to:

“Energy conservation is a hinge between descriptions, not a complete model of a process.”

Use all three cases. Include the terms *system boundary*, *store*, *transfer*, *sign convention*, and *empirical closure*. Explain why $\sum \Delta V = 0$, $\Delta U = Q - W$, and $E_{\text{initial}} = E_{\text{final}}$ express the same accounting discipline while referring to different physical content.

Teacher guide and indicative answers

The Energy Hinge

1. Kirchhoff’s second law

System and meaning. A convenient system is battery, resistors and connecting circuit. The battery raises electric potential energy per unit charge; the resistors transfer that energy to internal energy of the components and surroundings.

Kirchhoff’s loop rule gives

$$+12.0 - IR_1 - IR_2 = 0.$$

Therefore

$$I = \frac{12.0}{4.0 + 2.0} = 2.0 \text{ A},$$

$$V_1 = IR_1 = 8.0 \text{ V}, \quad V_2 = IR_2 = 4.0 \text{ V}.$$

The loop sum is $+12.0 - 8.0 - 4.0 = 0$.

For $q = 3.0 \text{ C}$,

$$E_{\text{battery}} = q\mathcal{E} = 36 \text{ J},$$

$$E_1 = qV_1 = 24 \text{ J}, \quad E_2 = qV_2 = 12 \text{ J}.$$

Conceptual answer. Kirchhoff’s second law enforces conservation of energy per unit charge around a closed loop. It does not supply the component relation $V = IR$, the battery’s emf, or the microscopic mechanism by which electrical energy becomes internal energy.

Diagnostic misconception. Voltage is often treated as something “used up”. The energy is transferred; voltage is the signed change in energy per coulomb.

2. First law of thermodynamics

Boundary and signs. The gas is the system. Heating transfers 500 J into it; expansion work transfers 200 J out.

$$\Delta U = Q - W = 500 - 200 = 300 \text{ J}.$$

As a zero-sum ledger,

$$Q - W - \Delta U = 0$$

or $+500 - 200 - 300 = 0$.

For a monatomic ideal gas,

$$\Delta T = \frac{2\Delta U}{3nR} = \frac{600}{3(0.20)(8.31)} \approx 120 \text{ K}.$$

Conceptual answer. The first law is universal bookkeeping. The relation $U = \frac{3}{2}nRT$ is model-specific closure for a monatomic ideal gas. It connects the stored internal energy to a measurable state variable. The first law alone does not give the final pressure, volume, path, rate of heat transfer or work mechanism.

Heat and work. They name modes of energy transfer across a boundary, not substances or stores contained by the gas. Once transferred, the energy contributes to the system’s internal energy and possibly other stores.

Diagnostic misconception. Students may treat Q as “heat inside the gas”. Keep the grammatical distinction: energy is transferred by heating or by work.

3. Nuclear binding energy

$$\Delta m = (1.007276 + 1.008665 - 2.013553) \text{ u} = 0.002388 \text{ u}.$$

Hence

$$E_{\text{rel}} = 0.002388(931.5) \approx 2.224 \text{ MeV}.$$

An appropriate ledger is

$$(m_p + m_n)c^2 = m_dc^2 + E_{\text{radiation}} + K_{\text{recoil}}.$$

The released-energy total is the sum of radiation and recoil kinetic energy.

System boundary. For the nucleons alone, energy leaves the system. For a larger closed system containing the deuteron and emitted radiation, total energy is constant. Rest mass is one contribution to total energy, so a lower final rest mass does not represent missing energy.

Binding energy. The bound state lies 2.224 MeV below the separated state. Binding energy is quoted positively as the minimum energy that must be supplied to separate the deuteron. Formation releases the same amount.

Conceptual answer. Conservation relates the measured masses to the released energy; it does not predict the deuteron’s mass or the strength of the nuclear bound state. Those are empirical inputs, or outputs of a deeper nuclear interaction model.

Diagnostic misconception. “Mass is converted into energy” can obscure that mass is already a form of system energy. The total mass-energy ledger remains balanced.

Synthesis and classroom use

Indicative synthesis.

All three cases impose a signed conservation ledger. Kirchhoff’s second law balances energy per unit charge around a loop; the first law balances transfers by heating and work against a change in internal energy; nuclear binding balances initial rest energy against final rest energy, radiation and recoil. The hinge is common, but the terms are not. Ohm’s law, a thermodynamic model of internal energy, and measured nuclear masses supply different forms of empirical closure. Energy conservation constrains any acceptable account; it does not identify the mechanism or determine every term.

Discussion prompts.

- What is the system in each case, and what changes when the boundary changes?
- Which terms are stores and which are transfers?
- Which sign convention is being used, and how could an alternative convention express the same conservation statement?
- Which relation closes each problem after conservation has been written?
- Why can a conservation law rule out an answer without being able to generate the answer by itself?

Suggested timing: 9 minutes per case, 10 minutes synthesis, 8 minutes whole-class comparison.